AWS LEARNING

Assignment No. 1

Cloud Computing Fundamentals & AWS Global Infrastructure

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Q1. What is Cloud Computing? Explain it in your own words and give two real-life examples of cloud usage.

Answer:

Cloud computing means using computing resources — servers, storage, databases, networking and software — over the internet from a cloud provider such as AWS, instead of buying, installing and maintaining that hardware ourselves. The provider owns the data centers; we simply rent whatever we need, whenever we need it, and pay only for what we actually use.

In simple words, it works like an electricity connection. We do not build a power plant at home — we just switch on the light and pay the bill for the units consumed. In the same way, we do not buy a physical server; we launch one on AWS in a few minutes, use it, and shut it down when the work is over.

Key characteristics: on-demand self-service, pay-as-you-go pricing, elasticity (scale up or down at any time), broad network access from any device, and no responsibility for the physical hardware.

Two real-life examples:

- **Google Drive / Google Photos —** our files, photos and documents are not stored on our own laptop but on Google's servers. We can open the same file from a phone, laptop or a cyber-cafe computer, and we never worry about disk failure or backups.
- **Netflix / Hotstar —** the movies and web series are stored and streamed from cloud servers (Netflix runs on AWS). Millions of people watch at the same time in the evening, and the platform automatically arranges extra servers to handle that rush.

Other everyday examples are Gmail, WhatsApp chat backup on Google Drive/iCloud, and online exam or classroom portals.

Q2. A company currently has its own physical servers, but the servers are expensive to maintain. They want to move their infrastructure to AWS. What advantages can they expect after moving to the cloud? Explain each advantage in short.

Answer:

After migrating from on-premises physical servers to AWS, the company can expect the following advantages:

- **No large upfront investment (CapEx → OpEx) —** there is no need to purchase servers, racks, UPS or cooling equipment. The heavy one-time capital expense becomes a small monthly operational expense.

- **Pay-as-you-go pricing —** the company is billed only for the resources actually used (per second/hour for EC2, per GB for S3). Idle hardware no longer wastes money.
- **Elasticity and scalability —** capacity can be increased during peak load and reduced afterwards using Auto Scaling. With physical servers, buying a new machine takes weeks; on AWS it takes minutes.
- **No hardware maintenance headache —** AWS takes care of power, cooling, physical security, hardware failure and replacement. The company's IT team can focus on the application instead of the data center.
- **High availability and fault tolerance —** the application can be deployed across multiple Availability Zones, so a single hardware or site failure does not bring the business down.
- **Reliable backup and disaster recovery —** automated snapshots, S3 versioning and cross-region copies give a low-cost DR setup that would be very expensive to build with a second physical site.
- **Speed and agility —** new environments for testing or a new product can be created in minutes, so features reach the market faster.
- **Global reach —** the application can be deployed in AWS Regions near the customers and cached at Edge Locations, which reduces latency worldwide.
- **Strong security and compliance —** IAM for fine-grained access control, encryption at rest and in transit, KMS, security groups and CloudTrail auditing, plus AWS's own compliance certifications, under the Shared Responsibility Model.
- **Managed services reduce workload —** services like Amazon RDS, S3, AWS Lambda and CloudWatch remove routine tasks such as patching, replication and monitoring setup.

Q3. Differentiate between IaaS, PaaS, and SaaS with one AWS example for each.

Answer:

Point	IaaS (Infrastructure as a Service)	PaaS (Platform as a Service)	SaaS (Software as a Service)
Meaning	The provider gives raw IT infrastructure — virtual servers, storage and network — which we configure ourselves.	The provider gives a ready platform to build, deploy and run applications; we only supply the code.	The provider gives complete ready-to-use software over the internet; we just log in and use it.
Managed by user	OS, runtime, middleware, application, data and patching inside the instance.	Only the application code and its data.	Nothing technical — only our own data and user settings.
Managed by AWS	Physical servers, storage, network and virtualisation.	Everything up to the runtime — OS, servers, scaling and load balancing.	The entire stack including the application itself.
Control vs ease	Maximum control, maximum responsibility.	Balanced — less control, much less maintenance.	Least control, easiest to use.
AWS example	**Amazon EC2 (virtual servers); also Amazon EBS and Amazon VPC.**	**AWS Elastic Beanstalk (also AWS Lambda and Amazon RDS).**	**Amazon WorkMail / Amazon WorkDocs / Amazon Chime.**
Typical user	System administrators and infrastructure engineers.	Application developers.	End users of the business.

In one line: IaaS gives us the land and bricks (EC2), PaaS gives us a ready building where we only arrange the furniture (Elastic Beanstalk), and SaaS gives us a fully furnished flat that we simply move into (Amazon WorkMail, or Gmail outside AWS).

Q4. A company wants to keep its critical applications on its own servers but use the cloud for additional resources when required. Which cloud deployment model would be most suitable and why?

Answer:

Most suitable model: Hybrid Cloud.

A hybrid cloud connects the company's existing private/on-premises data center with a public cloud such as AWS, so that both work together as one environment. The critical applications and sensitive data stay on the company's own servers, while AWS is used for extra capacity whenever demand increases — a pattern known as cloud bursting.

Why it is the best choice here:

- **Control over critical workloads —** the most important applications and confidential data remain inside the company's own data center, under its direct control.
- **Meets compliance and data-residency rules —** regulated data (financial, medical or government) that must not leave the premises can legally stay on-premises.
- **Cloud bursting for peak demand —** during a sale, month-end processing or a sudden traffic spike, extra servers are launched on AWS and released afterwards, so the company does not have to buy hardware for a few busy days.
- **Cost efficiency —** the existing investment in hardware keeps giving value, and the company pays AWS only for the temporary extra usage.
- **Better disaster recovery —** AWS can act as the backup/DR site for the on-premises workloads at a fraction of the cost of a second physical data center.
- **Smooth, gradual migration —** applications can be moved to the cloud one by one instead of attempting a risky big-bang migration.

AWS services that enable it: AWS Direct Connect or Site-to-Site VPN for private connectivity, AWS Outposts and VMware Cloud on AWS for AWS infrastructure on-premises, AWS Storage Gateway for hybrid storage, and AWS DataSync for data transfer.

Why not the others: a fully public cloud would mean moving the critical applications out of the company's own data center, and a fully private cloud would again require buying hardware for the peak load — which is exactly the cost problem they want to avoid.

Q5. Suppose you are using Blinkit/Swiggy/Amazon and thousands of users are placing orders at the same time. Why would such an application need cloud infrastructure?

Answer:

Such applications face traffic that is huge, sudden and unpredictable — for example a rush at lunch and dinner time, during an IPL match, on a rainy evening, or during a festival sale. Only cloud infrastructure can handle this pattern economically. The main reasons are:

- **Elasticity / Auto Scaling —** thousands of simultaneous orders need many servers, but only for a few hours. AWS Auto Scaling adds EC2 instances automatically when traffic rises and removes them when it falls.
- **Load balancing —** an Elastic Load Balancer spreads incoming requests across many servers in different Availability Zones, so no single server crashes under the load.
- **High availability —** even a few minutes of downtime during peak hours means lost orders and lost revenue, so the app must run in multiple AZs with automatic failover.
- **Handling peaks without wasting money —** buying physical servers for the busiest two hours of the year would leave them idle the rest of the time; pay-as-you-go avoids that.
- **Databases that can scale —** managed services such as Amazon RDS with read replicas or Amazon DynamoDB handle millions of reads and writes for orders, carts and inventory.
- **Large media storage and fast delivery —** lakhs of product images and restaurant photos are kept in Amazon S3 and delivered quickly through CloudFront Edge Locations.
- **Real-time features —** live order tracking, delivery-partner location updates and push notifications need always-available, low-latency backend services.
- **Data analytics and personalisation —** recommendations, demand prediction, dark-store stock planning and surge pricing need big-data processing power on demand.
- **Multi-city / global reach —** servers close to the users in each region give faster app response and a better ordering experience.
- **Security of payments —** encryption, IAM, AWS WAF and DDoS protection (AWS Shield) protect payment and personal data even during heavy traffic.

Q6. Explain the difference between an AWS Region and an Availability Zone (AZ). Why does AWS have multiple Availability Zones within a Region?

Answer:

Point	AWS Region	Availability Zone (AZ)
Definition	A separate geographic area of the world where AWS has a cluster of data centers.	One or more discrete data centers inside a Region, each with independent power, cooling and networking.
Example	Mumbai (ap-south-1), N. Virginia (us-east-1), Frankfurt (eu-central-1).	ap-south-1a, ap-south-1b, ap-south-1c inside the Mumbai Region.
Size / scope	Contains a minimum of two — usually three or more — Availability Zones.	Contains one or more physical data centers.
Distance	Regions are hundreds or thousands of kilometres apart.	AZs in a Region are a few kilometres apart — far enough to avoid a common disaster, close enough for low-latency links.
Purpose	Chosen for latency to users, data-residency laws, service availability and price.	Used to build high availability and fault tolerance inside the chosen Region.
Connectivity	Regions are connected over the AWS global backbone network.	AZs are connected by high-bandwidth, low-latency, redundant private fibre links.

Why AWS keeps multiple Availability Zones in one Region:

- **Fault isolation —** each AZ has its own power, cooling and network, so a failure in one AZ (power cut, fire, flood or hardware fault) does not affect the others.
- **High availability —** an application spread over two or three AZs keeps running even if one complete AZ goes down.
- **Disaster resilience —** AZs are physically separated by a meaningful distance, so a local disaster cannot destroy all of them together.
- **Synchronous replication with low latency —** the fast private links between AZs allow services like Amazon RDS Multi-AZ to keep a real-time standby copy of the database.
- **Scalability and capacity —** more AZs mean more physical capacity, so customers can launch large workloads inside the same Region.
- **Meeting SLAs —** the AWS availability SLA for services such as Amazon EC2 assumes the workload is deployed across multiple AZs.

Q7. A company hosts its website in Mumbai, but customers from the USA and Europe are accessing it. How can AWS Edge Locations help improve the user experience?

Answer:

If the servers are only in the Mumbai Region, every request from a user in the USA or Europe has to travel thousands of kilometres and back. This adds high latency, so pages and images load slowly. AWS Edge Locations — used by Amazon CloudFront (CDN), Amazon Route 53 and AWS Global Accelerator — solve exactly this problem.

How Edge Locations improve the experience:

- **Content is cached near the user —** static content such as images, CSS, JavaScript and video is copied to Edge Locations in New York, London, Frankfurt and so on, and the user is served from the nearest edge instead of from Mumbai.
- **Much lower latency —** the round trip becomes a few milliseconds instead of hundreds of milliseconds, so the website feels fast and responsive.
- **Faster dynamic content too —** requests that must reach the Mumbai origin travel over the optimised AWS backbone network instead of the public internet, which is quicker and more stable.
- **Reduced load on the origin server —** most repeated requests are answered by the cache, so the Mumbai servers handle far less traffic and can run on smaller instances.
- **Lower data-transfer cost —** serving from the cache is cheaper than serving every byte from the origin Region.
- **Better availability —** if the origin is briefly unreachable, cached content can still be served to users.
- **Security at the edge —** AWS Shield (DDoS protection), AWS WAF and HTTPS/TLS termination happen at the Edge Location, so attacks are stopped far away from the origin.
- **Smart routing —** Amazon Route 53 latency-based or geolocation routing sends each user to the fastest available endpoint.

Result: customers in the USA and Europe get almost the same fast experience as customers in India, without the company building any servers outside Mumbai.

Q8. Imagine an application is running only in one Availability Zone. That AZ suddenly becomes unavailable. What problem could occur, and how can the company design the application to improve availability?

Answer:

Problems that could occur:

- **Complete outage —** the whole application becomes unreachable, because a single AZ is a single point of failure. This is 100% downtime, not just a slowdown.
- **Possible data loss —** if the database and its EBS volumes/snapshots exist only in that AZ, recent data may be lost or locked until the AZ recovers.
- **Business and revenue loss —** orders, payments and transactions stop; for an e-commerce or banking application even a few minutes are very costly.
- **Loss of customer trust and reputation —** users move to a competitor's app and the brand is damaged publicly.
- **SLA breach and penalties —** the company may fail the uptime it promised to its own customers.
- **Long, manual recovery —** without a standby setup the team must rebuild servers and restore backups by hand, which takes hours.

How to design for better availability:

- **Deploy across multiple Availability Zones —** run EC2 instances in at least two or three AZs of the Region instead of one.
- **Use an Elastic Load Balancer —** an ALB/NLB spanning several AZs performs health checks and automatically sends traffic only to healthy instances in healthy AZs.
- **Use an Auto Scaling group across AZs —** it maintains the desired number of instances and relaunches capacity in a healthy AZ if one AZ fails.
- **Enable Amazon RDS Multi-AZ (or Amazon Aurora) —** a synchronous standby database in another AZ takes over automatically during a failure.
- **Store objects in Amazon S3 —** S3 already replicates data across multiple AZs automatically, so files and images stay safe.
- **Make the application stateless —** keep sessions in Amazon ElastiCache or DynamoDB instead of on the local server, so any instance in any AZ can serve any user.
- **Add Route 53 health checks and DNS failover —** traffic is redirected away from an unhealthy endpoint automatically.
- **Plan multi-Region disaster recovery —** for very critical systems keep backups, AMIs and cross-region replication so the workload can be restored in another Region.
- **Use Infrastructure as Code and test failover —** CloudFormation/Terraform plus periodic failover drills ensure the environment can be recreated quickly and that the design actually works.

Q9. Choose any one application that you use regularly — such as Instagram, Netflix, WhatsApp, Amazon, Blinkit or Swiggy — and identify: a) Why it needs cloud computing b) What type of data it may store c) Why high availability is important for it.

Answer:

Application chosen: Instagram

a) Why Instagram needs cloud computing

- **Massive global scale —** more than two billion users around the world upload and view content every day; no fixed-size company-owned data center could serve this.
- **Unpredictable traffic spikes —** a celebrity post, a festival, New Year's Eve or a cricket final creates sudden bursts of traffic that need instant extra capacity.
- **Huge storage requirement —** photos, videos, Reels and Stories add petabytes of data continuously; cloud object storage grows without any hardware planning.
- **Fast global delivery —** a CDN with edge caching is needed so a Reel loads instantly in India, the USA and Europe alike.
- **Heavy computing for AI/ML —** feed ranking, the Explore page, Reels recommendations, face/object detection and content moderation need large-scale GPU and compute power on demand.
- **Cost efficiency and development speed —** pay-as-you-go resources and managed services let new features be tested and released quickly.

b) What type of data it may store

- **User account data —** name, username, email, phone number, date of birth, bio, profile photo and hashed passwords.
- **Media content —** photos, videos, Reels, Stories, highlights and their thumbnails in multiple resolutions.
- **Social graph data —** followers, following, close-friends lists and blocked accounts.
- **Engagement data —** likes, comments, shares, saves, views and watch time.
- **Messages —** direct messages, media shared in chats and message metadata.
- **Metadata —** timestamps, captions, hashtags, tagged users, device details and optional location/geotags.
- **Behavioural and analytics data —** search history, scrolling behaviour, session logs and interests used for recommendations.
- **Advertising and business data —** ad campaigns, impressions, clicks, insights for creator and business accounts, and payment details for ads.

c) Why high availability is important for it

- **24 × 7 global usage —** users are active in every time zone, so there is never a safe window for the service to be down.
- **Direct revenue impact —** advertising is the main income; every minute of downtime means lost impressions and lost advertiser money.
- **Livelihood of creators and businesses —** influencers and small shops depend on Instagram for reach and sales, so an outage directly affects their income.
- **Real-time features —** DMs, Live video and Stories must work instantly; delays or failures break the core experience.
- **Reputation and user trust —** an outage immediately trends on other platforms and pushes users towards competitors.
- **Data safety —** users' memories are stored there, so permanent loss of photos would be unacceptable — data must be replicated across AZs and Regions.

Q10. Draw a simple diagram of AWS Global Infrastructure showing the relationship between Region → Availability Zones → Data Centers → Edge Locations. Also add 3–4 lines explaining how these components work together.

Answer:

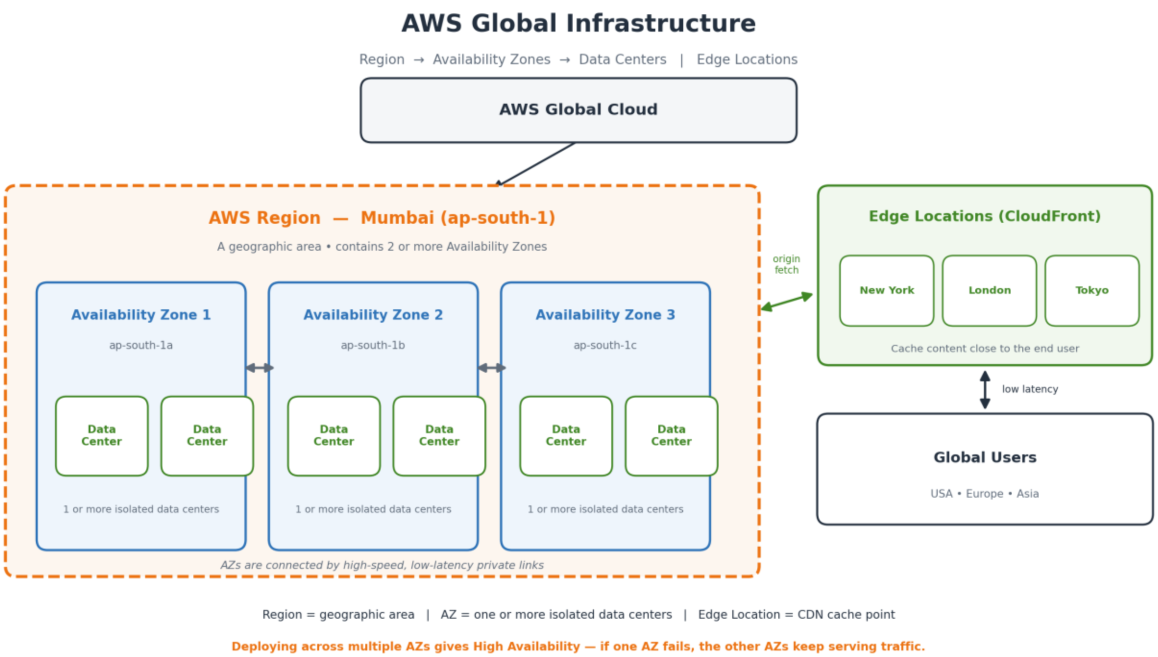

Figure 1: AWS Global Infrastructure — Region → Availability Zones → Data Centers, with Edge Locations serving users globally.

Text form of the hierarchy:

```
AWS Global Cloud
  |
  +-- Region  (e.g. Mumbai / ap-south-1)
  |      +-- Availability Zone 1 (ap-south-1a)  -->  Data Center 1, Data Center 2
  |      +-- Availability Zone 2 (ap-south-1b)  -->  Data Center 1, Data Center 2
  |      +-- Availability Zone 3 (ap-south-1c)  -->  Data Center 1, Data Center 2
  |
  +-- Edge Locations  (New York, London, Tokyo, Mumbai, ...)  -->  End Users
```

How these components work together:

- A Region is a large geographic area where AWS has built its infrastructure, and every Region contains at least two — usually three or more — Availability Zones.
- Each Availability Zone is made up of one or more physically separate data centers with their own power, cooling and networking, and the AZs are joined by high-speed, low-latency private links.
- When an application is deployed in two or three AZs of the same Region, the failure of one AZ (or of one data center inside it) does not stop the application — this is how high availability and fault tolerance are achieved.

- Edge Locations are a much larger, separate network of small sites spread across many cities. Through Amazon CloudFront they cache content coming from the Region and deliver it from the point nearest to the user, which reduces latency and takes load off the origin servers.
- Together they give the best of both: the Region and its AZs provide reliable compute and storage, while the Edge Locations provide fast global delivery to end users.

Conclusion

This assignment covered the meaning and benefits of cloud computing, the three service models (IaaS, PaaS and SaaS), the hybrid deployment model, and the building blocks of the AWS Global Infrastructure — Regions, Availability Zones, Data Centers and Edge Locations. The main takeaway is that the cloud removes the cost and the limits of owning hardware, and that designing across multiple Availability Zones, with Edge Locations for global reach, is what makes an application scalable, fast and highly available.